



A special thanks to

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Massive Near-Field Spatial Multiplexing:

Higher Capacity Without More Bandwidth

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1.

**Network capacity
and signal dimensions**



2.

**Massive near-field
spatial multiplexing**



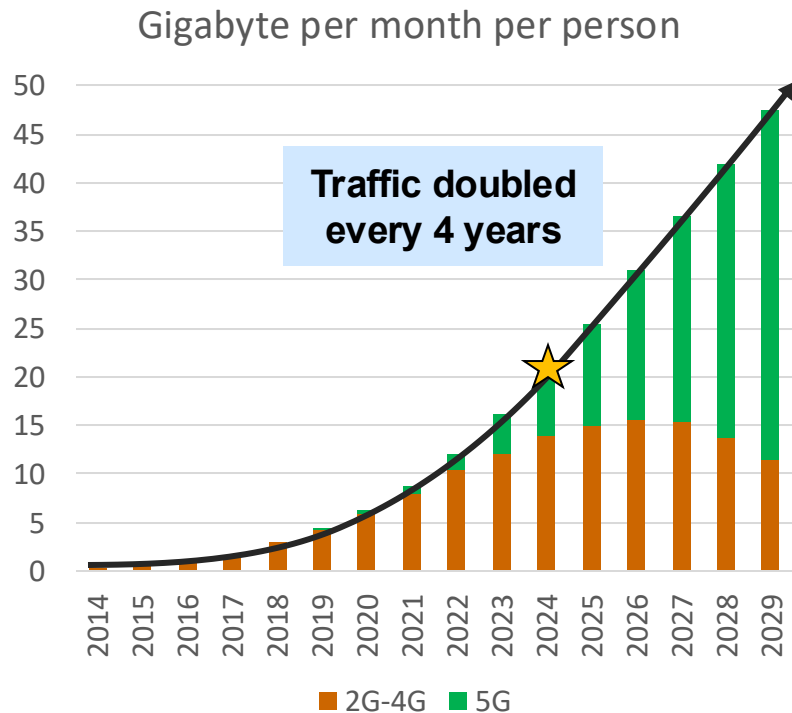
3.

**What might
happen in 6G?**



Network Capacity in Mobile Networks

Demand



Ericsson Mobility Report, 2024

Supply

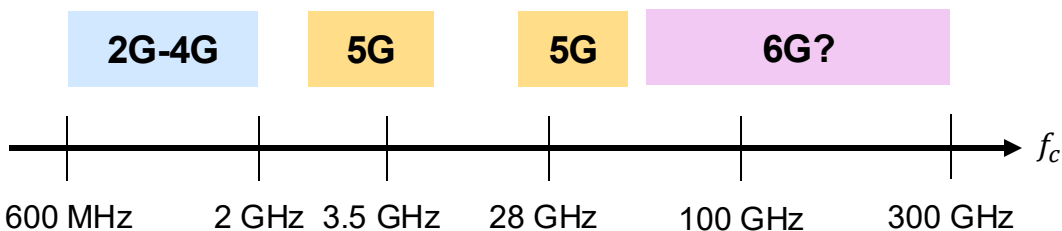
Channel capacity (bit/s per access point):

$$C = \text{Bandwidth} \cdot \text{Layers} \cdot \log_2(1 + \text{SNR})$$

We need more bandwidth (Hz)?

The rise of mmWave...

Rule-of-thumb: Bandwidth $\propto f_c$



...and fall?

South Korea cancels SKT's 28 GHz 5G licence

Written by [Mary Lennighan](#) 15 May 2023 @ 12:38



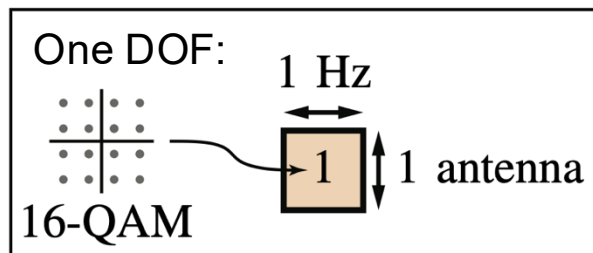
South Korea has withdrawn SK Telecom's licence to operate 5G services in the 28 GHz band, the telco having failed to meet its rollout requirements.

What Really Matters: Degrees-of-Freedom (DOF)

“Building Blocks of Wireless Signals”

- **Bit rate formula:**

$$\text{bit/s} = \text{bit/DOF} \cdot \text{DOF/s}$$



5G today

($f_c = 3.5$ GHz, $B = 100$ MHz)

8 spatial DOF

100 million spectral DOFs

10 bit/DOF (1024-QAM)

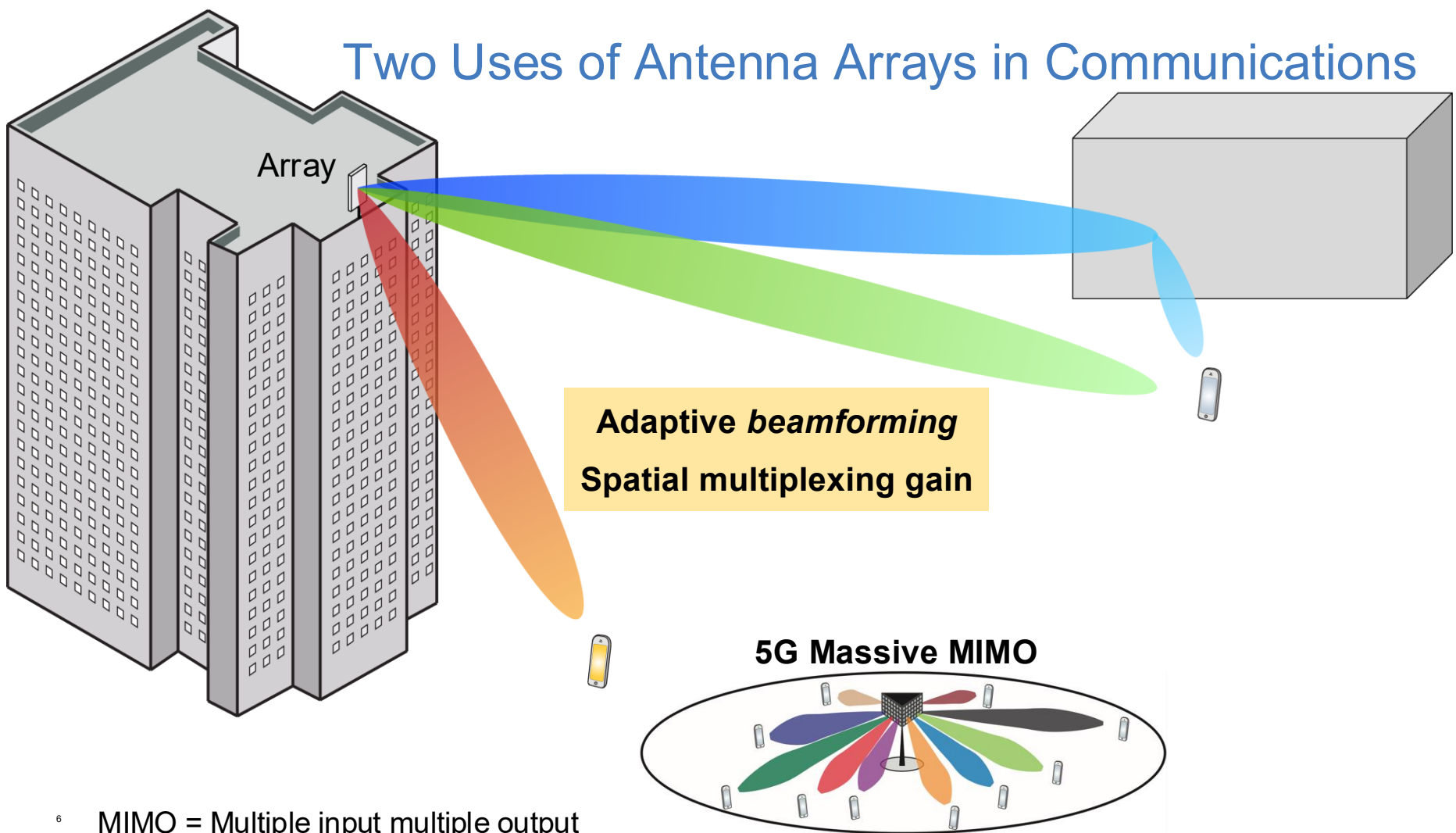
Theoretically up to 8 Gbps

Spatial DOF

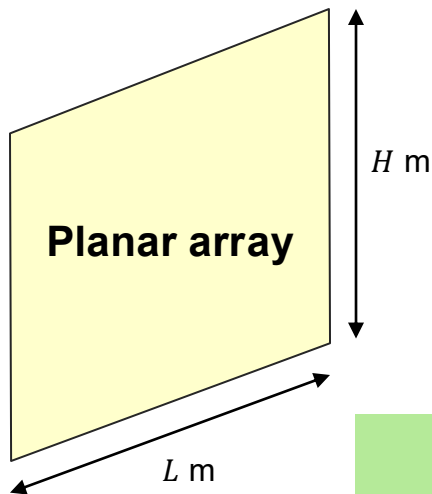
Can we expand the
spatial DOFs in the future?

Spectral DOF

Two Uses of Antenna Arrays in Communications



Quantifying the Theoretical Spatial DOFs



Maximum value

$$\text{DOF} \approx \pi \frac{LH}{\lambda^2} \text{ spatial streams}$$

Reference:

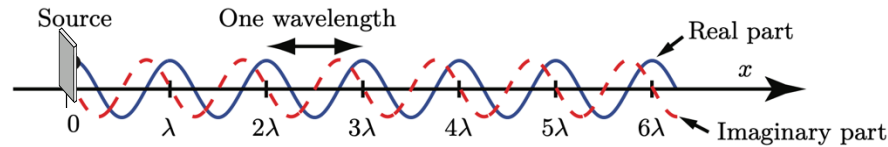
A. S. Y. Poon, R. W. Brodersen, and D. N. C. Tse, "Degrees of freedom in multiple-antenna channels: A signal space approach", IEEE Trans. Inf. Theory, 2005.

S. Hu, F. Rusek, O. Edfors, "Beyond Massive MIMO: The Potential of Data Transmission with Large Intelligent Surfaces," IEEE Trans. Signal Process., 2018.

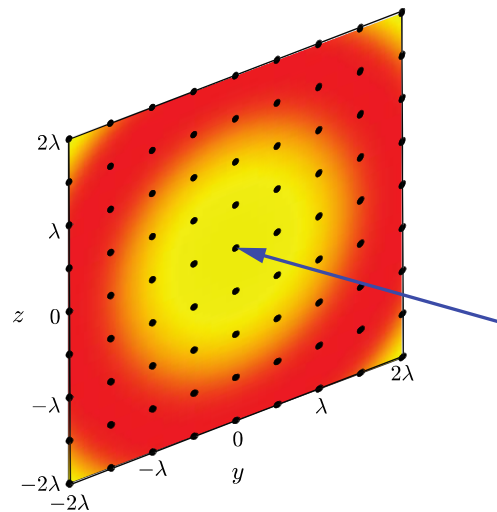
Uplink: The Array Samples the Impinging Waveform

Signal: $s \cdot e^{-j2\pi x/\lambda}$

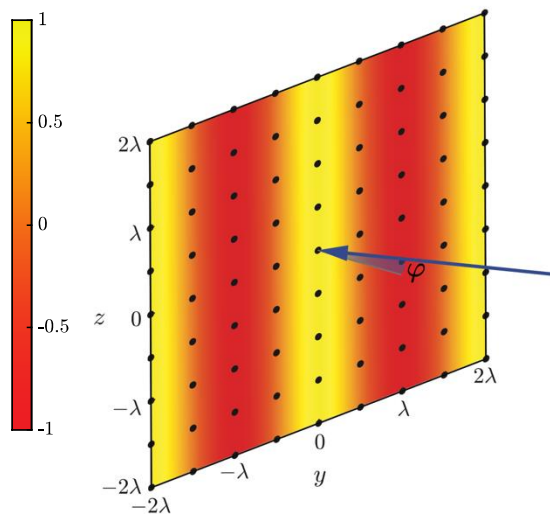
↑
Data



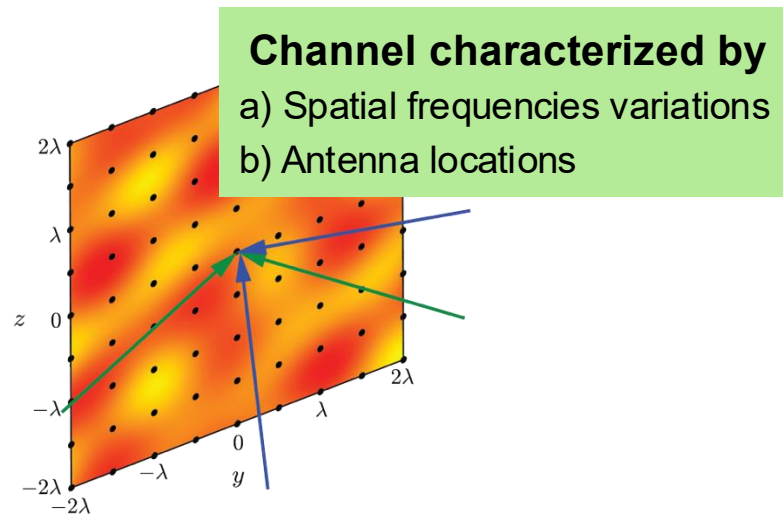
Spatial frequency: $\frac{1}{\lambda}$



User 1: near array in
broadside ($d = 4\lambda$)



User 2: far from array,
 $\phi = 30^\circ$ azimuth angle



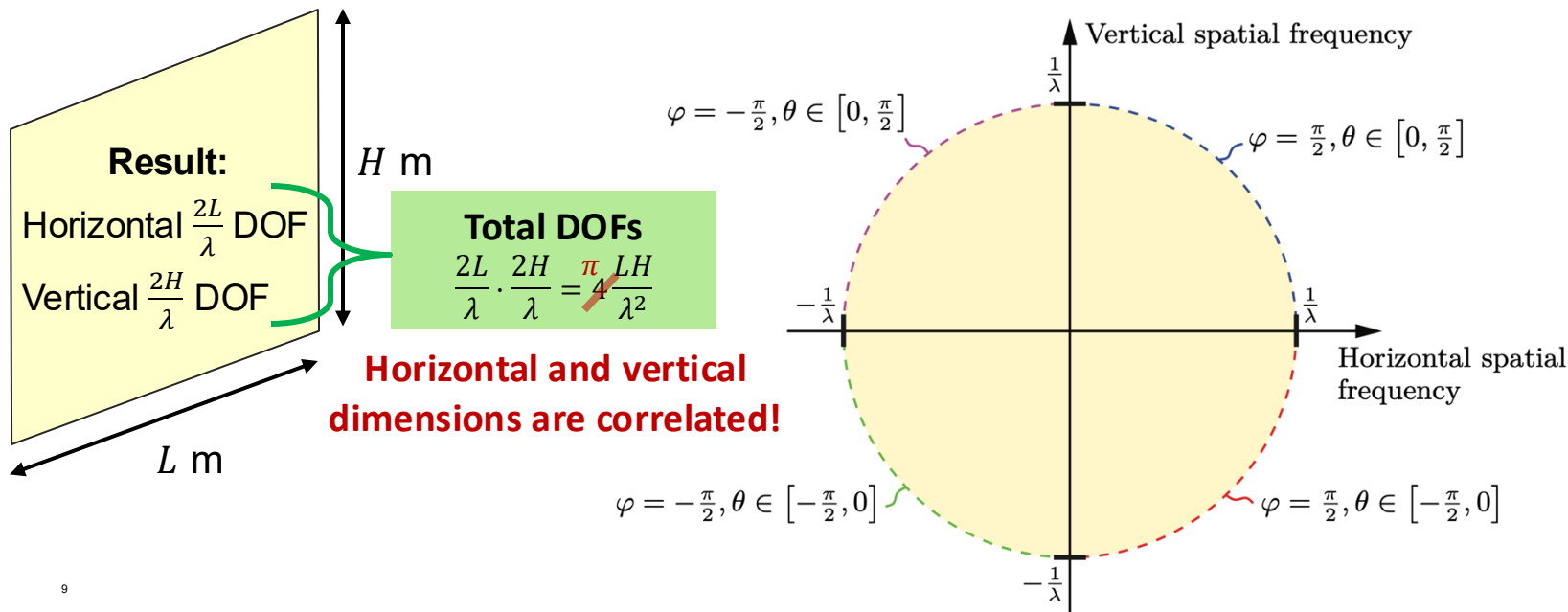
User 3: four far-field
propagation paths

Nyquist Sampling Theorem

- A complex-valued continuous-time signal ~~$g(t)$~~ $g(x)$ ^{space}
- Only contains frequencies in an interval smaller than ~~B~~ Hz
 - Entirely determined by samples spaced ~~$1/B$~~ seconds apart

Signal described by ~~B~~ DOF/s ^{$2/\lambda$ DOF/m}

$\lambda/2$ meters



What are the Implications?

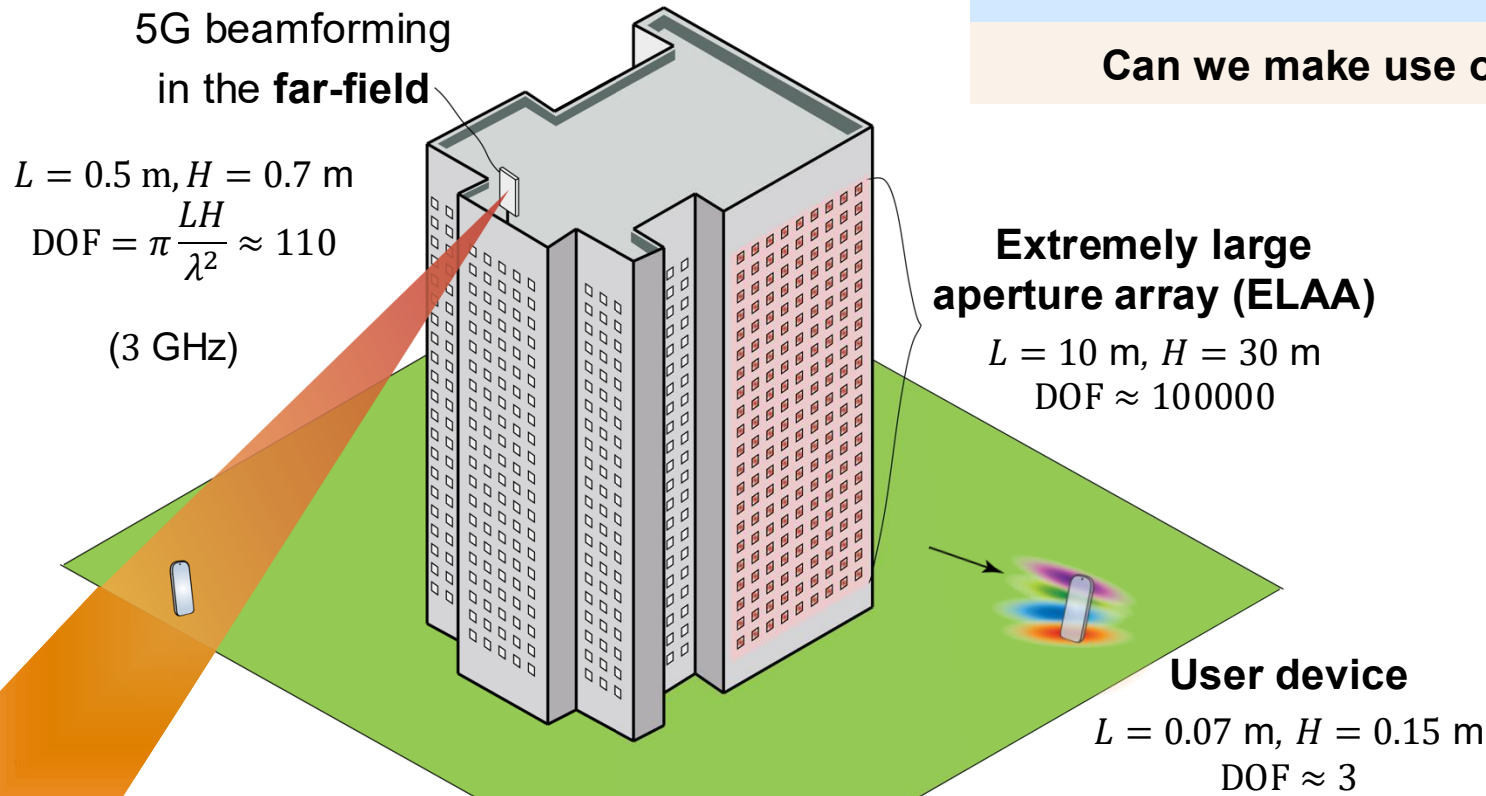
Can we get even more spatial DOFs?

Dual polarization: $2 \times$ DOFs

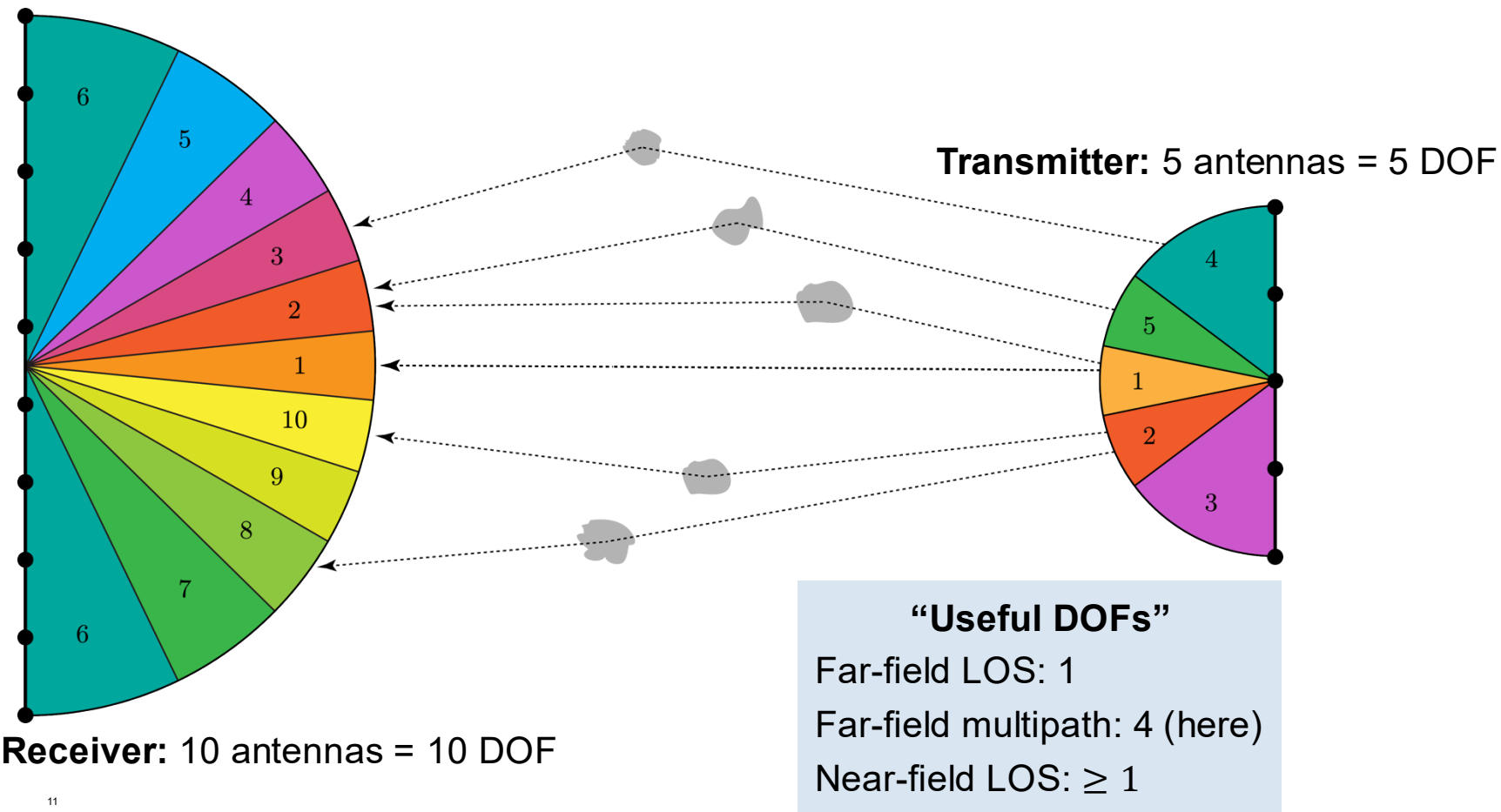
mmWave (3 $\rightarrow$ 30 GHz):

100 $\times$ more DOFs

Can we make use of them all?

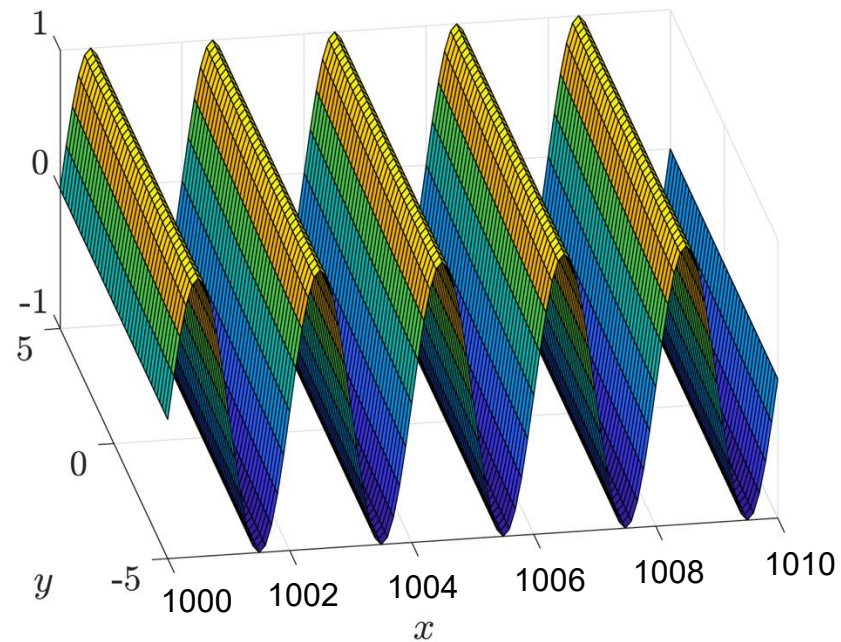
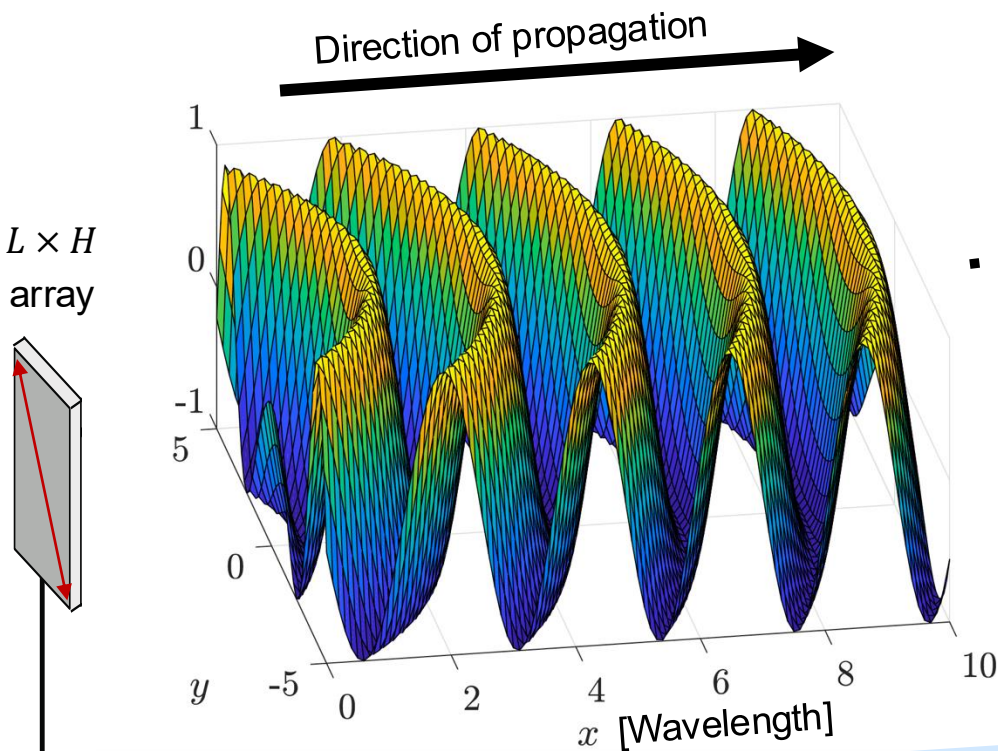


Useful DOFs Require Transmitter-Receiver Pairing



MASSIVE NEAR-FIELD SPATIAL MULTIPLEXING

From Spherical Waves to Approximately Planar Waves



Reactive
near-field

Radiative near-field

User distance $< 2 \cdot \frac{L^2 + H^2}{\lambda}$

Fraunhofer
distance

Far-field regime
Traditional scenario

Spatial Multiplexing in Both Angle and Depth

5G array:



Fraunhofer
↑

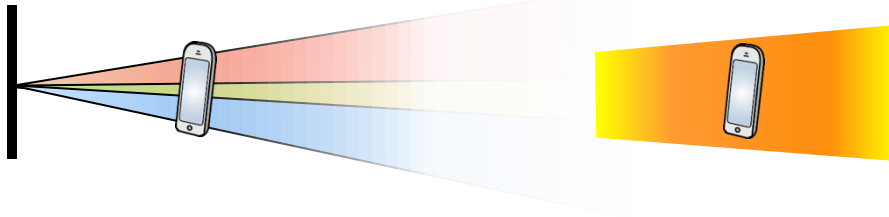
Example: Fraunhofer distance

1×1 m, 3 GHz: 40 m

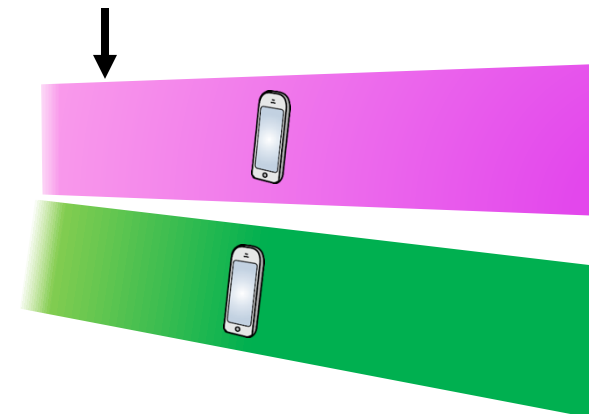
1×1 m, 30 GHz: 400 m

10×10 m, 3 GHz: 4 km

ELAA:



Fraunhofer
↓



Larger antenna array

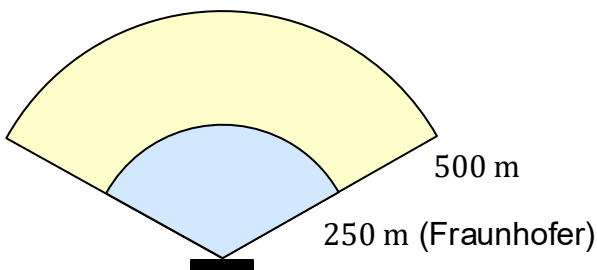
Narrower beams and finite depth in radiative near-field

Exploiting Depth for Spatial Multiplexing of Many Users

Base station with N antennas

Channel vector, user k : $\mathbf{h}_k \in \mathcal{S} \subset \mathbb{C}^N$

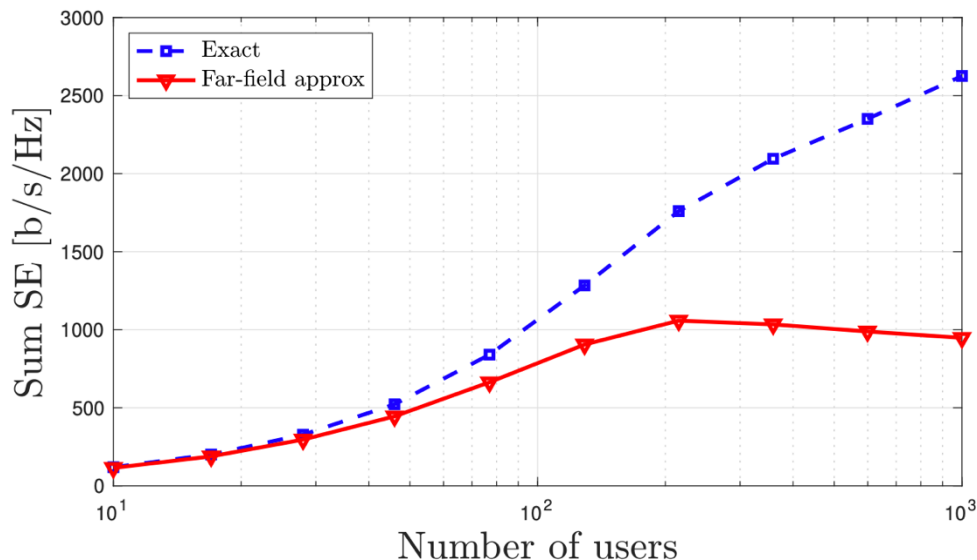
Subset of physically possible vectors
(Dimension = DOF)



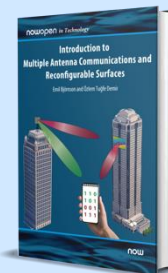
$N = 5000$ antennas
30 GHz, 1×0.5 m

All N -dimensional vectors

“Richer” channels
in the near-field



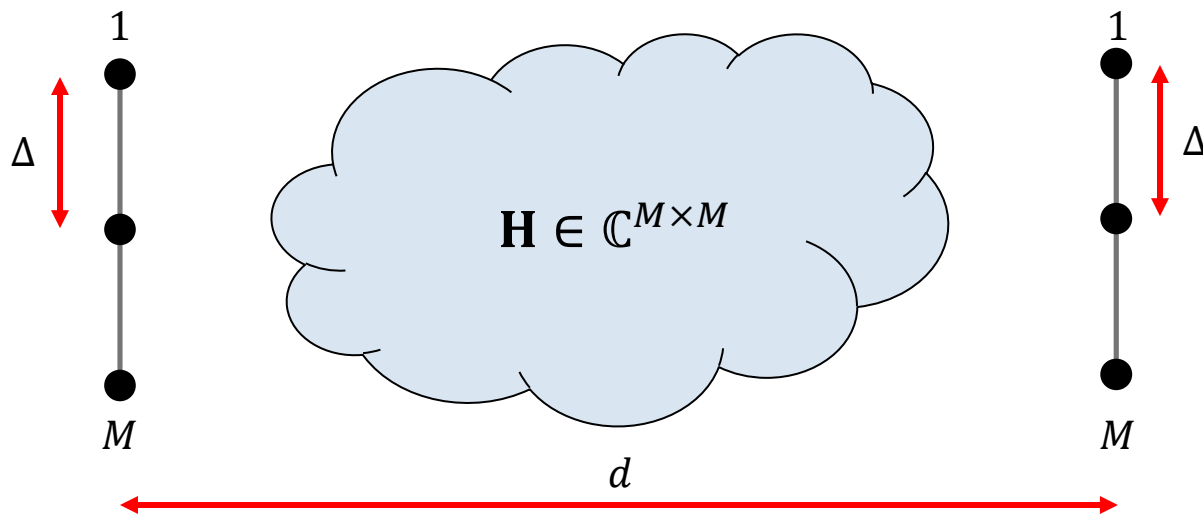
Classical
algorithms



“Towards 6G MIMO: Massive Spatial Multiplexing, Dense Arrays, and Interplay Between Electromagnetics and Processing,” arXiv:2401.02844

Many DOF per Device: Line-of-Sight (LOS) Capacity Maximization

MIMO = Multiple Input Multiple Output

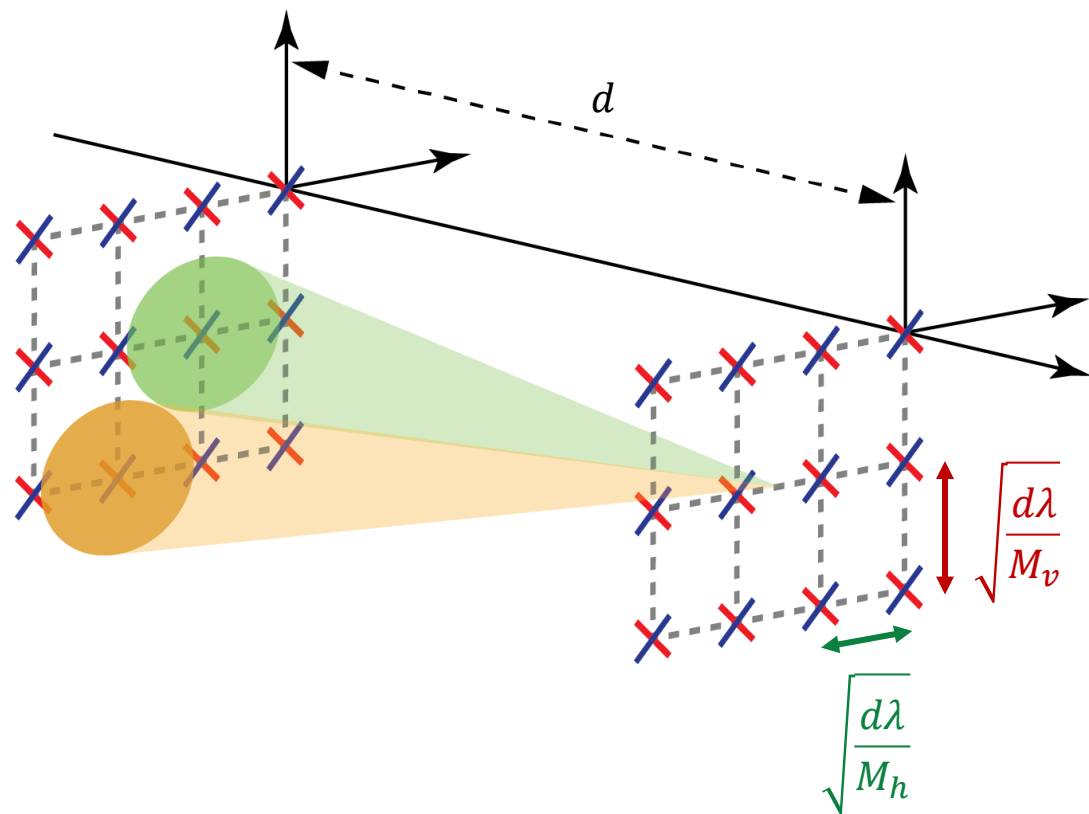


Problem: Optimize spacing Δ to maximize MIMO capacity

High SNR: M equal singular values

Solution: Apply parabolic approximation of spherical waves
Enforce that the columns of $\mathbf{H}$ are orthogonal

Optimized Planar Dual-Polarized $M_h \times M_v$ Arrays



$$\text{Area} = M_h \sqrt{\frac{d\lambda}{M_h}} M_v \sqrt{\frac{d\lambda}{M_v}} = d\lambda \sqrt{M}$$

with $M = M_h M_v$

Number of antennas in a fixed area:

$$M = \left(\frac{\text{Area}}{d\lambda} \right)^2$$

$2M$ DOFs with equal singular values
(Value independent of M)

Fraction of maximum DOF

$$\frac{\left(\frac{\text{Area}}{d\lambda} \right)^2}{\pi \frac{\text{Area}}{\lambda^2}} = \frac{\text{Area}}{\pi d^2} \ll 1$$

Reference: A. Irshad, A. Kosasih, E. Björnson, L. Sanguinetti, "Optimal Dual-Polarized Arrays for Massive Capacity Over Point-to-Point MIMO Channels," TWC 2024

Scaling Law: Channel Capacity vs. Wavelength

Capacity formula [bit/s]:

$$C = \underset{\substack{\uparrow \\ \text{Bandwidth}}}{B} \cdot \underset{\substack{\uparrow \\ \text{Spatial DOFs}}}{2M} \cdot \log_2 \left(1 + \frac{\underset{\substack{\downarrow \\ \text{Received power per antenna}}}{P_r}}{\underset{\substack{\uparrow \\ \text{Noise power}}}{2BN_0}} \right)$$

$$M = \left(\frac{\text{Area}}{d\lambda} \right)^2$$

Isotropic antennas, area $A = \frac{\lambda^2}{4\pi}$

$$P_r = \frac{\lambda^2}{(4\pi d)^2} \cdot P_t$$

$$C \rightarrow \left(\frac{\text{Area}}{4\pi d^2} \right)^2 \frac{P_t}{N_0} \log_2(e) = \text{constant as } \lambda \rightarrow 0$$

Directive antennas

Antenna areas A_t and A_r with $A_r A_t \propto \lambda^{4-\rho}$

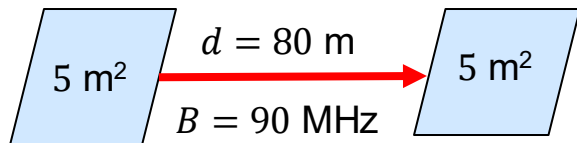
$$P_r = \frac{A_r A_t}{d^2 \lambda^2} \cdot P_t$$

$$C = O\left(\frac{1}{\lambda^{4-\rho}}\right) \rightarrow \infty \text{ as } \lambda \rightarrow 0 \text{ if } \rho \in (0, 2]$$

Great Capacity Without More Bandwidth

Reference: A. Irshad, A. Kosasih, E. Björnson, L. Sanguinetti, "Optimal Dual-Polarized Arrays for Massive Capacity Over Point-to-Point MIMO Channels," TWC 2024

Line-of-sight scenario:



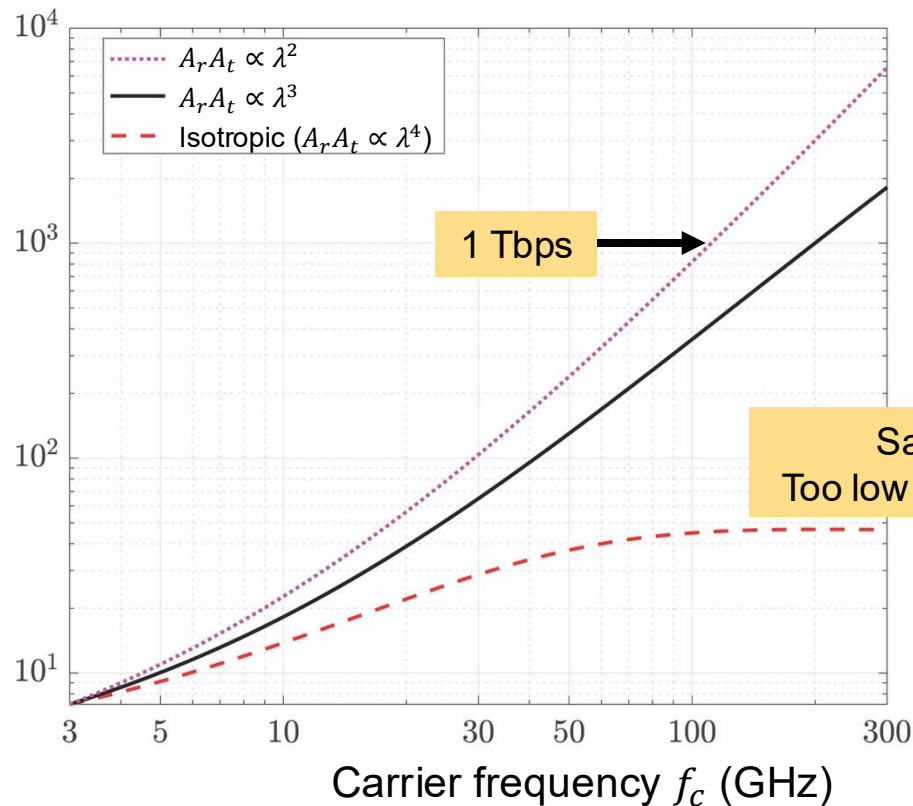
Slightly directive antennas

Area: $A_r = A_t \propto \lambda$, Gain: $\propto 1/\lambda$

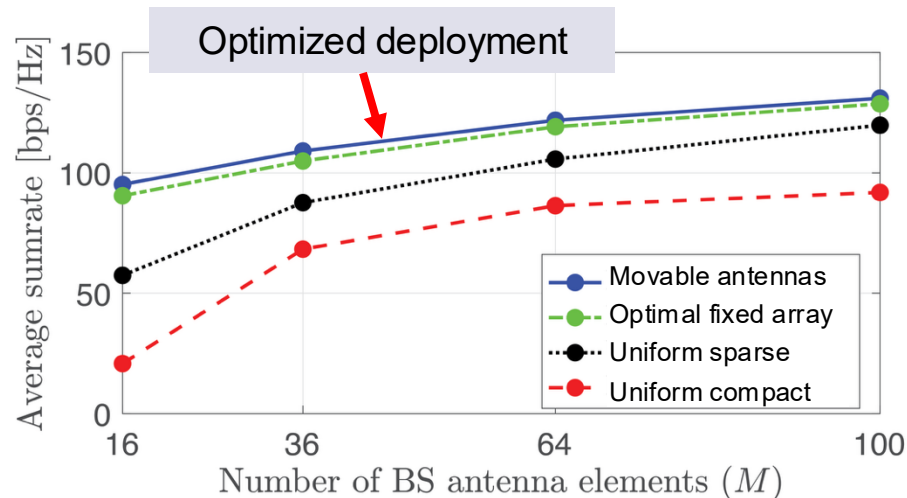
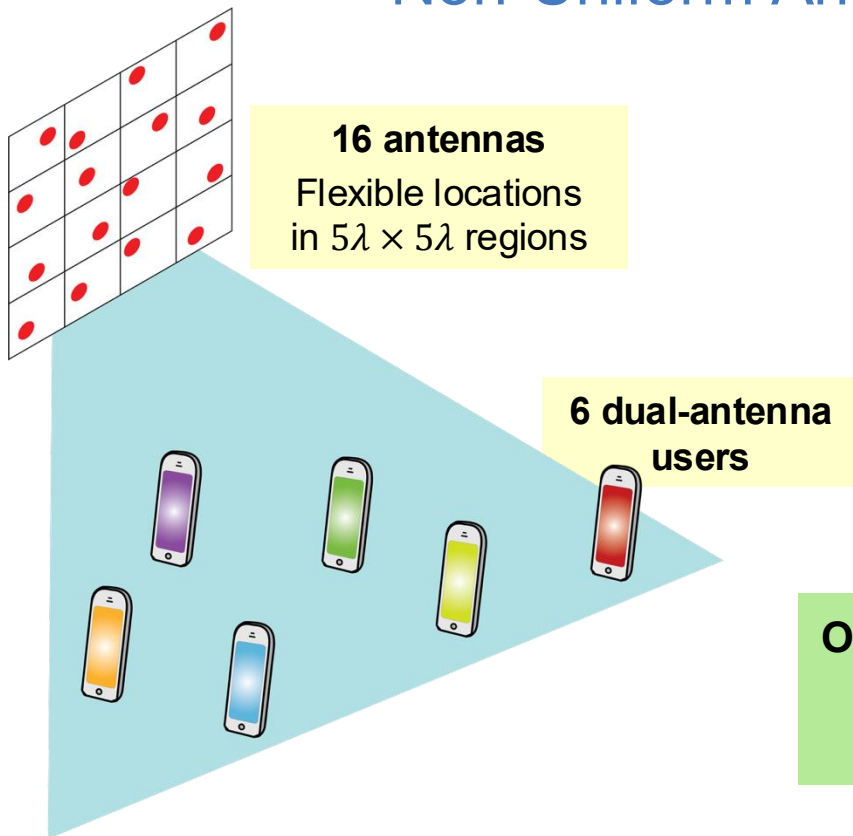
Spatial DOFs: $\left(\frac{\text{Area}}{d\lambda}\right)^2 = O(f_c^2)$

Capacity: $C = O(f_c^2)$

Capacity (Gbit/s)



Non-Uniform Arrays: Are They Useful? Yes!



Optimized for the coverage area
Sparse instead of dense
Limited need for movable antennas

Reference: A. Irshad, A. Kosasih, V. Petrov, E. Björnson,
"Pre-Optimized Irregular Arrays versus Movable
Antennas in Multi-User MIMO System," IEEE WCOML, 2025

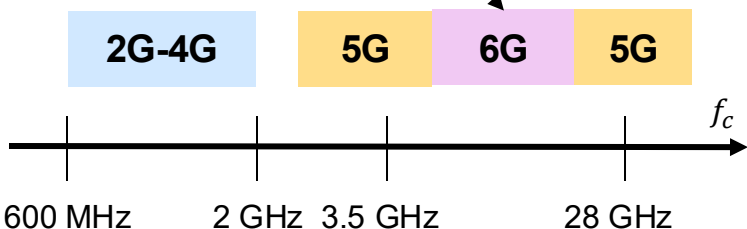
WHAT MIGHT HAPPEN IN 6G?

6G Frequency Bands...

...and Implications

Frequency range 3 (FR3)

- 7.1-8.4 GHz
- 14.8-15.35 GHz



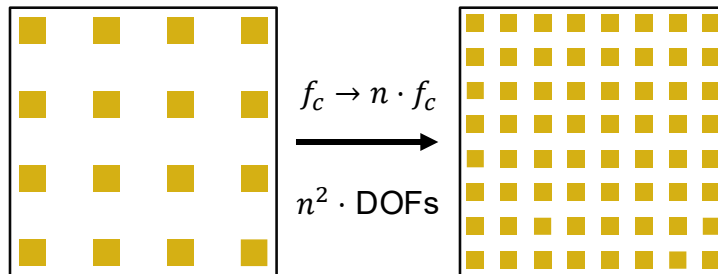
Massive MIMO

16, 32 or 64 antennas

Gigantic MIMO

256, 512, ...

Number of spatial DOFs



Frequency band

- 7.1-8.4 GHz
- 14.8-15.35 GHz

DOF scaling

$$n^2 = 5$$
$$n^2 = 18$$

More DOFs in 6G!

2-3× more spectral DOFs

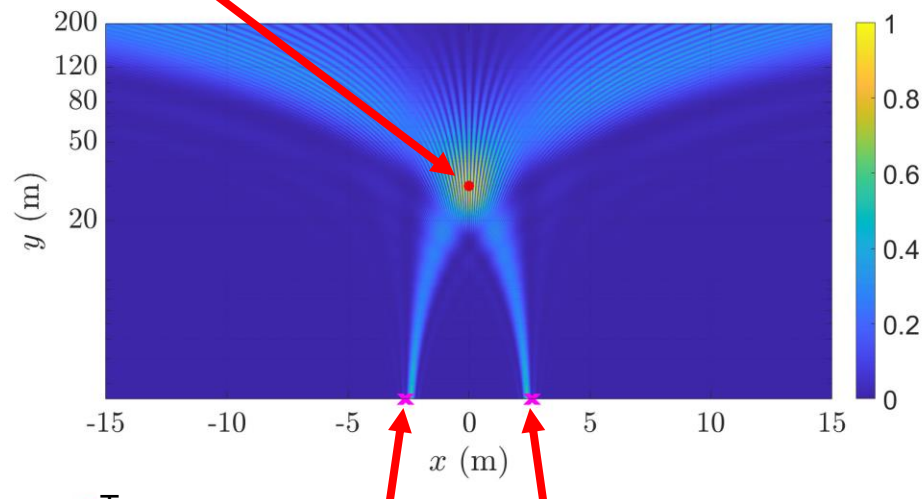
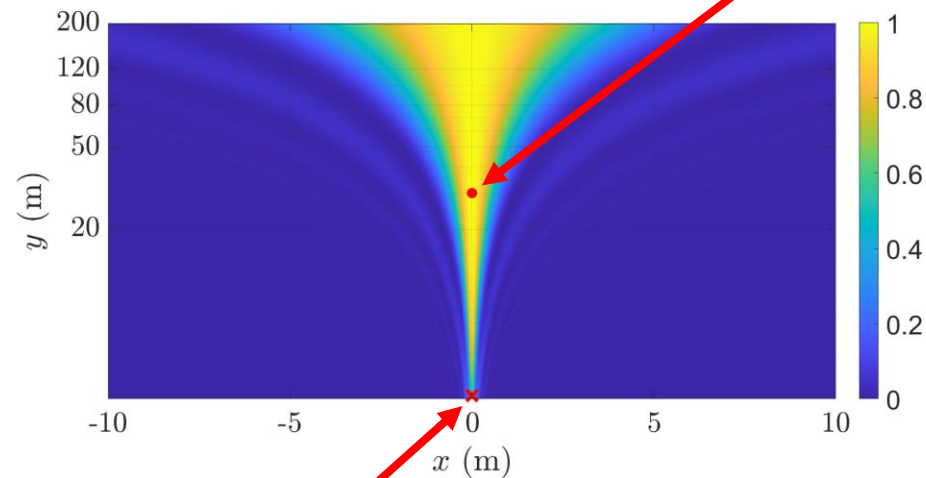
5-18× more spatial DOFs



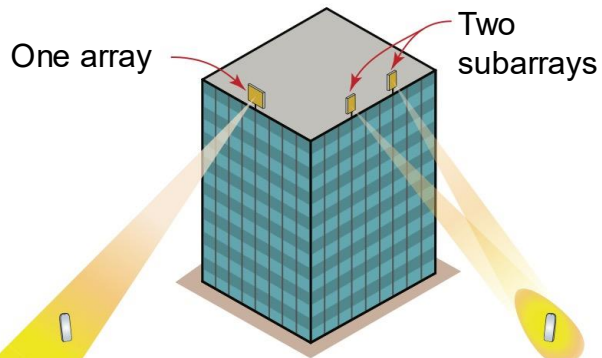
Will Near-Field Effects Appear in 6G?

Yes, with distributed subarrays

Receiver at 30 m



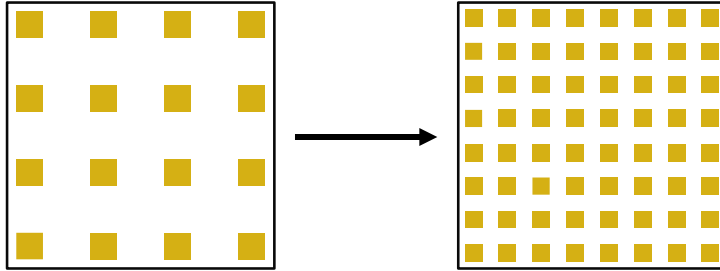
Two 24-antenna ULAs, 5 m apart



Uniform linear array (ULA)
48 antennas (0.5 m at 15 GHz)
Fraunhofer distance: 23 m

Summary

Much Higher Capacity in 6G Without More Bandwidth

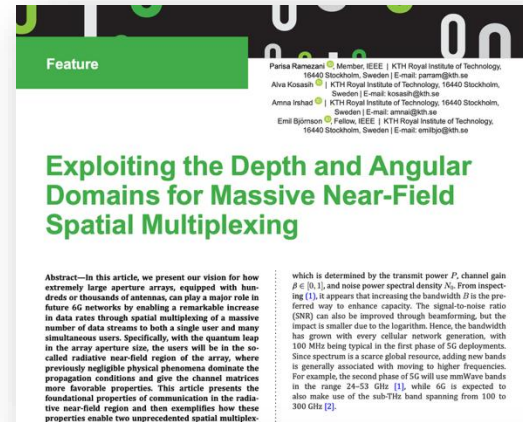


Capacity grows as f_c^2 thanks to MIMO

- Faster than $O(f_c)$ with spectrum (Key in 6G!)
- Maximum DOFs **and** practically useful DOFs
- Array design essential to maintain the SNR

Near-field propagation effects

- Richer channels: Control both angle and depth
- Achievable with “Gigantic MIMO” and subarrays



Enabling 6G Performance in the Upper Mid-Band by Transitioning From Massive to Gigantic MIMO

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ABSTRACT The initial 6G networks will likely operate in the upper mid-band (7–24 GHz), which has decent propagation conditions but underwhelming new spectrum availability. In this paper, we explore whether we can anyway reach the ambitious 6G performance goals by evolving the multiple-input multiple-output (MIMO) technology from massive in 5G to gigantic in 6G. We describe how many antennas are needed to reach the envisioned 6G peak user rates, how many can realistically be deployed in practical radio equipment, and what the practical spatial degrees-of-freedom might become. We further suggest a new deployment strategy that enables the utilization of radiative near-field effects in these bands for precise beamforming, localization, and sensing from a single base station site. Finally, we identify open research and standardization challenges that must be overcome to efficiently use gigantic MIMO dimensions in 6G from hardware, cost, and algorithmic perspectives.

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